```
#include <SoftwareSerial.h>

#include "VoiceRecognitionV3.h"

#include <LiquidCrystal.h>

LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

VR myVR(A1,A0);    // 2:RX 3:TX, you can choose your favourite pins.

int relayPins[4] = {9, 10, 11, 12};

uint8_t record[7]; // save record

uint8_t buf[64];

bool relayState[4] = {LOW, LOW, LOW, LOW};

bool trainingMode = false;

int led = 13;


int group = 0;


#define switchRecord        (0)


#define group0Record1       (1)

#define group0Record2       (2)

#define group0Record3       (3)

#define group0Record4       (4)

#define group0Record5       (5)

#define group0Record6       (6)


#define group1Record1       (7)

#define group1Record2       (8)

#define group1Record3       (9)

#define group1Record4       (10)

#define group1Record5       (11)
```

```
#define group1Record6      (12)

void setup()
{
  /** initialize */
  myVR.begin(9600);
  Serial.begin(115200);
  Serial.println("Elechouse Voice Recognition V3 Module\r\nMulti Commands sample");
  pinMode(led, OUTPUT);
  for (int i = 0; i < 4; i++) {
  pinMode(relayPins[i], OUTPUT);
  }
  for (int i = 0; i < 4; i++) {
  digitalWrite(relayPins[i], LOW);
  }
  lcd.begin(16, 2);
  lcd.setCursor(0, 0);
  lcd.print("Voice System");
  delay(1500);
  if(myVR.clear() == 0){
    Serial.println("Recognizer cleared.");
    lcd.setCursor(0, 1);
    lcd.print("Recognizer cleared.");
  }else{
    Serial.println("Not find VoiceRecognitionModule.");
    Serial.println("Please check connection and restart Arduino.");
    lcd.setCursor(0, 1);
    lcd.print("Please check");
```

```
    while(1);
  }

  record[0] = switchRecord;
  record[1] = group0Record1;
  record[2] = group0Record2;
  record[3] = group0Record3;
  record[4] = group0Record4;
  record[5] = group0Record5;
  record[6] = group0Record6;
  group = 0;
  if(myVR.load(record, 7) >= 0){
    printRecord(record, 7);
    Serial.println(F("loaded."));
    lcd.clear();
    lcd.setCursor(0, 1);
    lcd.print("loaded  ");
  }
  lcd.setCursor(0, 0);
  lcd.print("D0  D1  D2  D3  ");
  lcd.setCursor(0, 1);
  lcd.print("OFF OFF OFF OFF ");
}

void loop()
{
  int ret;
  ret = myVR.recognize(buf, 50);
```

```
if(ret>0){
  Serial.print("rxd=");
  Serial.println(buf[1]);
  if(buf[1] == 0)
  {
   Serial.println("light on");
   digitalWrite(9, HIGH);
   lcd.setCursor(0, 1);
   lcd.print("ON  ");

  }
  if(buf[1] == 1)
  {
   Serial.println("light off");
   digitalWrite(9, LOW);
   lcd.setCursor(0, 1);
   lcd.print("OFF ");
  }
  if(buf[1] == 2)
  {
   Serial.println("fan on");
   digitalWrite(10, HIGH);
   lcd.setCursor(4, 1);
   lcd.print("ON  ");
  }
  if(buf[1] == 3)
  {
   Serial.println("fan off");
```

```
 digitalWrite(10, LOW);
 lcd.setCursor(4, 1);
 lcd.print("OFF ");
}
if(buf[1] == 4)
{
 Serial.println("tv on");
 digitalWrite(11, HIGH);
 lcd.setCursor(8, 1);
 lcd.print("ON  ");
}
if(buf[1] == 5)
{
 Serial.println("tv off");
 digitalWrite(11, LOW);
 lcd.setCursor(8, 1);
 lcd.print("OFF ");
}
if(buf[1] == 6)
{
 Serial.println("ac on");
 digitalWrite(12, HIGH);
 lcd.setCursor(12, 1);
 lcd.print("ON  ");
}
if(buf[1] == 7)
{
 Serial.println("ac off");
```

```
 digitalWrite(12, LOW);
 lcd.setCursor(12, 1);
 lcd.print("OFF ");
}
switch(buf[1]){
  case switchRecord:
    /** turn on LED */
    if(digitalRead(led) == HIGH){
      digitalWrite(led, LOW);
    }else{
      digitalWrite(led, HIGH);
    }
    if(group == 0){
      group = 1;
      myVR.clear();
      record[0] = switchRecord;
      record[1] = group1Record1;
      record[2] = group1Record2;
      record[3] = group1Record3;
      record[4] = group1Record4;
      record[5] = group1Record5;
      record[6] = group1Record6;
      if(myVR.load(record, 7) >= 0){
        printRecord(record, 7);
        Serial.println(F("loaded."));
      }
    }else{
      group = 0;
```

```
        myVR.clear();
        record[0] = switchRecord;
        record[1] = group0Record1;
        record[2] = group0Record2;
        record[3] = group0Record3;
        record[4] = group0Record4;
        record[5] = group0Record5;
        record[6] = group0Record6;
        if(myVR.load(record, 7) >= 0){
          printRecord(record, 7);
          Serial.println(F("loaded."));
        }
      }
      break;
    default:
      break;
  }
  /** voice recognized */
  printVR(buf);
 }
}

/**
  @brief   Print signature, if the character is invisible,
           print hexible value instead.
  @param   buf     --> command length
           len     --> number of parameters
*/
```

```
void printSignature(uint8_t *buf, int len)
{
  int i;
  for(i=0; i<len; i++){
    if(buf[i]>0x19 && buf[i]<0x7F){
      Serial.write(buf[i]);
    }
    else{
      Serial.print("[");
      Serial.print(buf[i], HEX);
      Serial.print("]");
    }
  }
}

/**
  @brief   Print signature, if the character is invisible,
           print hexible value instead.
  @param   buf  -->  VR module return value when voice is recognized.
             buf[0]  -->  Group mode(FF: None Group, 0x8n: User, 0x0n:System
             buf[1]  -->  number of record which is recognized.
             buf[2]  -->  Recognizer index(position) value of the recognized record.
             buf[3]  -->  Signature length
             buf[4]~buf[n] --> Signature
*/
void printVR(uint8_t *buf)
{
  Serial.println("VR Index\tGroup\tRecordNum\tSignature");
```

```
  Serial.print(buf[2], DEC);
  Serial.print("\t\t");

  if(buf[0] == 0xFF){
    Serial.print("NONE");
  }
  else if(buf[0]&0x80){
    Serial.print("UG ");
    Serial.print(buf[0]&(~0x80), DEC);
  }
  else{
    Serial.print("SG ");
    Serial.print(buf[0], DEC);
  }
  Serial.print("\t");

  Serial.print(buf[1], DEC);
  Serial.print("\t\t");
  if(buf[3]>0){
    printSignature(buf+4, buf[3]);
  }
  else{
    Serial.print("NONE");
  }
//  Serial.println("\r\n");
  Serial.println();
}
```

```
void printRecord(uint8_t *buf, uint8_t len)
{
  Serial.print(F("Record: "));
  for(int i=0; i<len; i++){
    Serial.print(buf[i], DEC);
    Serial.print(", ");
  }
}
```